

Artificial Intelligence Lab (AI-1002)

Mid Term Lab Exam

Total Time (mins): 100

Total Weightage: 25

Total Marks: 50

Total Questions: 03

Paper Type: A

Date: Wednesday, April 01st, 2026

Time: 01:40 PM to 03:20 PM

Lab Instructor(s):

Mr. Abdullah Riaz Shaikh

Mr. Abdullah Yaqoob

Student Name

Roll No

Section

Student Signature

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Attempt all the questions!

Important Instructions:

- Please submit the .ipynb file with the file name renamed as AI_Mid_Term_Lab_Exam_Your-Name_Your-Roll-No.ipynb.
- For Question 03 (initial code on Genetic Algorithm), get the code from the local submission folder by pressing Windows + R and entering \\172.16.5.43\Exam Material\Abdullah Shaikh\Q3.py.

LLO # 01: Explain and implement BFS and DFS algorithms, comparing their efficiency and applications in uninformed search.

Question # 01:

[Weightage: 07 | Marks: 14 | Time: 25 minutes]

A hospital uses an autonomous medical drone to deliver critical medicines to patients across different wards. The hospital is represented as a graph of rooms/wards, each connected by corridors. Some rooms have patients in critical condition, and some have normal patients or are empty.

The drone has the following goals:

1. Deliver medicines to critical patients first (higher utility).
2. Minimize time/moves to maximize efficiency (higher utility = critical patients treated quickly with fewer moves).

The drone can:

1. Move to an adjacent room (cost = 1 move).
2. Deliver medicine if the patient is critical (increase utility by 10 points per critical patient treated).
3. Skip rooms that have no critical patients (to save moves).

Environment Representation

We represent the hospital as a dictionary (graph), like this:

```
hospital = {  
  
    "WardA": {"status": "Critical", "adjacent": ["WardB", "WardC"]},  
  
    "WardB": {"status": "Normal", "adjacent": ["WardA", "WardD"]},  
  
    "WardC": {"status": "Critical", "adjacent": ["WardA", "WardD"]},  
  
    "WardD": {"status": "Empty", "adjacent": ["WardB", "WardC"]},  
  
}
```

You are required to perform the following tasks:

1. Create a UtilityBasedAgent class to track the hospital environment, current ward, number of moves, and critical patients treated, and implement a utility() function as: $(10 \times \text{critical patients treated} - \text{moves})$. **[03 Marks]**
2. Implement a BFS function to find the shortest path from the current ward to the nearest ward with a critical patient. **[07 Marks]**
3. Define agent actions so that it delivers medicine if the current ward is critical; otherwise, it moves along the BFS path to the next critical ward while updating moves and environment state. **[02 Marks]**
4. Run a simulation until all critical patients are treated, printing the current ward, action taken, and utility after each step, and display the final utility at the end. **[02 Marks]**

LLO # 02: Apply and implement A* and GBFS algorithms, evaluating their heuristics, efficiency, and applications in informed search.

Question # 02:

[Weightage: 09 | Marks: 18 | Time: 35 minutes]

After a major earthquake, a city has been divided into a 2D grid map where each cell represents a location. A rescue drone must navigate the area to reach survivors while avoiding dangerous zones.

Grid Representation:

1. The environment is an $N \times M$ grid.
2. Each cell may contain:
 - $S \rightarrow$ Starting position of the drone
 - $G \rightarrow$ Goal (location of a survivor)
 - $\cdot \rightarrow$ Safe path (cost = 1)
 - $\# \rightarrow$ Blocked area (cannot pass)
 - $W \rightarrow$ Water/flooded area (cost = 3)
 - $F \rightarrow$ Fire zone (cost = 5)

The drone must find the optimal path from S to G such that the total cost is minimized.

	Col 0	Col 1	Col 2	Col 3	Col 4
Row 0	S	.	.	W	.
Row 1	.	#	.	#	.
Row 2	.	#	.	.	.
Row 3	.	.	F	#	G
Row 4	.	.	.	.	.

You are required to perform the following tasks:

1. Model the problem by defining the state space (all grid positions), initial state (S), goal state (G), allowed actions (up, down, left, right), and path cost based on terrain. **[05 Marks]**
2. Apply A* search using the evaluation function ($f(n) = g(n) + h(n)$) with Manhattan Distance as the heuristic. **[09 Marks]**
3. Show the sequence of nodes expanded during the execution of the algorithm, provide the optimal path from S to G, and Calculate the total cost of the final path. **[04 Marks]**

LLO # 04: Implement Genetic Algorithms to solve optimization problems, analyzing their operators, convergence, and performance.

Question # 03:

[Weightage: 09 | Marks: 18 | Time: 40 minutes]

A delivery company must visit N cities exactly once and return to the starting city. The cities are located on a 2D map with given coordinates. The company wants to minimize the total travel distance while visiting all cities.

Design a Genetic Algorithm to find an optimal or near-optimal route that:

1. Visits each of the N cities exactly once.
2. Returns to the starting city after visiting all cities.
3. Minimizes the total travel distance of the route.

$$\text{TotalDistance} = \sum_{k=0}^{N-1} d(c_k, c_{k+1})$$

You are provided with initial code. You are required to:

1. Calculate total distance for selection. [Formula Given] **[02 Marks]**
2. Use Tournament Selection for parent selection. **[03 Marks]**
3. Use Order Crossover (OX) for crossover. **[04 Marks]**
4. For mutation, swap two cities randomly. **[03 Marks]**
5. Stop when a predefined number of generations is reached, or when a route with desired minimal distance is found. **[06 Marks]**

Input:

1. Number of cities N.
2. Coordinates of each city (x_i, y_i) for i = 0 to N-1.

Output:

1. A route (ordered list of cities) showing the sequence in which to visit the cities.
2. Total distance of the route.

Input (5 cities):

City 0: (2,3)

City 1: (5,7)

City 2: (1,9)

City 3: (8,2)

City 4: (4,5)

Output:

Best Route: [1, 2, 0, 3, 4]

Total Distance: 23.87 units

Must return your paper after completion!

Best of luck!